

RESEARCH ARTICLE

Bioseparations and Downstream Processing

Comprehensive studies on building a scalable downstream process for mRNAs to enable mRNA therapeutics

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Funding information

Defense Advanced Research Projects Agency,
Grant/Award Number: HR011-18-3-001;
DARPA

Abstract

In recent years, mRNA-based therapeutics have been a fast-growing new class of biologics that can, in principle, encode any protein(s) directly in patients to treat various diseases. mRNA vaccines have been proven to work efficiently, have high potency, and can be rapidly developed and deployed, which is critical for a quick responses in the case of a pandemic. Such agile development is enabled by rapid synthesis of RNA in vitro using recombinant enzymes rather than relying on lengthy and complex cell culture processes. mRNA exhibits physical and chemical properties differing from protein-based therapeutics. It is highly negatively charged and the hydroxyl group makes mRNA less stable and more susceptible to hydrolysis and nucleophilic cleavage. This novel work shares comprehensive studies carried out to compare the performance of various mRNA purification strategies by considering its scalability and critical quality attributes. In addition, the paper provides insights on how to establish a scalable mRNA purification process that consists of ultrafiltration/diafiltration and chromatography steps with good recoveries. Alternative Oligo(dT) based columns were further explored aiming to improve total process recovery. With Oligo(dT) as a capture step, overall recoveries of 70% can be achieved for mRNAs studied here that encode anti-influenza immunoglobulin G monoclonal antibodies.

KEYWORDS

chromatography, dsRNA, manufacturing process, mRNA, TFF

1 | INTRODUCTION

In recent years, messenger RNA (mRNA) has emerged as a promising class of therapeutics to treat various diseases including cancers,¹ cystic fibrosis,² and metabolic liver disorder arginase deficiency.³ Recent advances in delivering an inhaled version of mRNA into lung epithelium demonstrated its great potential to treat lung diseases.^{4,5} Increasing attention has been paid to novel mRNA vaccines owing to their high potency, capacity for rapid development as well as their simplified manufacturing procedures and safe administration.^{6,7} While two

COVID19 mRNA vaccines have been approved from Pfizer-BioNTech and Moderna, many others are in clinical and preclinical trials.⁸ mRNA is also widely used in the areas of genome editing (CRISPR-Cas9) and gene replacement,⁹ which offers great opportunity in the gene and cell therapy area.^{10,11}

mRNA vaccines exhibit advantages and drawbacks over traditional and current generation vaccines. Compared with Adeno associated virus (AAV) and plasmid DNA (pDNA) based therapeutics that often must breach the nucleus for therapeutic effect, mRNA must overcome the lower bar of entering the cytoplasm. While AAV and pDNA medicines offer the potential for long-lasting or curative effects, they can come with concerns over limited repeated dosing or

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genome integration respectively.¹² In contrast, its flexibility for repeated dosing or multivalent dosing favors the mRNA drug format in addressing the pandemic with the likely need for variant associated protection through mono or multivalent boosters.^{13,14} However, the changing vaccine landscape has seen biopharmaceutical companies take different approaches to address the COVID19 pandemic, including mRNA (Pfizer-BioNTech and Moderna), AAV(Oxford- AstraZeneca, CanSino, Janssen and Gamaleya center) as well as pDNA (Cadila Healthcare).

Genetic engineering approaches pairing the substitution of non-natural nucleotides and 5' cap analogues with sequence optimization or wholesale substitution of the mRNA elements (i.e., 5'—and 3'—UTRs, poly-adenosine [polyA] tail, T7 promoter, and primary mRNA sequence) have enabled efficient translation and increased half-life while substantially evading a host immune response.^{15,16} Moreover, clearance of immunogenic double stranded (ds) RNA has been of great interest and concern and its removal from mRNA drug substance has been substantially addressed.^{17–19} Importantly, recent work has significantly extended mRNA response durability by utilizing novel mRNA constructs,²⁰ extending this modality's promise for applications in protein replacement.

From a manufacturing perspective, major potential advantages arise from replacing mAb production with production of mRNA encoding that same protein. Contrasted with mAbs processed in a weeks-long culture of mammalian cells with a panoply of chemically diverse host-cell proteins²¹ and potential for adventitious virus,²² mRNA is synthesized in an in-vitro transcription reaction without live cells, contains fewer than 15 reagents, and is complete within less than half a day.^{23,24}

In order to advance commercial therapies, mRNA pharmaceuticals must overcome several generic and modality specific hurdles. Compared with mAbs, manufacturing of mRNA is likely better suited to templated platform purification processes owing to their decreased chemical heterogeneity.²⁵ In principle, chemical homogeneity can narrow the process design space, fitting better to quality-by-design principles that can hasten time to clinic by reducing the onerous tailoring of a platform purification process to a given molecule.²⁶ Still outstanding is a full understanding of how mRNA transcript length impacts bio-processes (i.e., chromatography, filtration operating limits) beyond the length constraints imposed by the formulation vehicle.²⁷ Ultimately, the process should also be flexible enough to cope with a diversity of mRNA lengths and sequences resulting in pure product, substantially clear of immunogenic species, such as dsRNA,²⁸ residual reagents, endotoxin, and bioburden.

While applications for therapeutic mRNA broaden, researchers are innovating ways to purify mRNA to improve productivity and cost of goods (COGs). For example, both strong and weak anion exchange chromatography (AEX) have been widely explored to purify mRNA from crude IVT reaction^{29,30} while mixed-mode core-bead chromatography,³¹ hydroxyapatite resins,³² temperature denaturing chromatography³³; paired H-bonding chromatography³⁴; tangential flow filtration (TFF)-mediated mRNA precipitation³⁵ have also been explored. More recently, vendors have successfully commercialized a

coupled Oligo(dT) oligonucleotide onto highly porous or convective monolith scaffolds to create mRNA affinity resins which were explored for mRNA purification of varying lengths.

The aim of this article is to provide practical process development insights focused on designing a suitable and scalable process by exploring a subset of chromatography (CHT type II, CaptoCore400® and Oligo(dT)) and TFF operations applied to paired mRNA molecules corresponding to the heavy chain (HC) and light chain (LC) of two distinct anti-influenza immunoglobulin G (IgG) molecules mAb1 and mAb2.

2 | MATERIALS AND METHODS

2.1 | Linearized DNA purification

In order to advance the most suitable scale-up process for mRNA production encoding mAb 1 and mAb2, we have also considered purification of DNA template. Plasmid DNA originating from *E. coli* was purified using a Giga prep kit (Qiagen) or equivalent in house protocol (not published) before type IIS restriction enzyme digestion resulting in a linear template. Linearized DNA were purified using five different methodologies prior to mRNA synthesis either with TFF or the combination of chromatography (bind and elute or flow through) and TFF followed by concentration and diafiltration. The quality of mRNA was analyzed and compared post in vitro transcription using capillary electrophoresis. Q Sepharose HP and PlasmidSelectXtra (Cytvia) resins were used during scouting experiments. For subsequent TFF, a 30KDa molecular weight cut off (MWCO) at regenerated cellulose membrane (MERCK) or equivalent was used.

2.2 | In vitro transcription

The mRNA encoding HC and LC were synthesized via a HiScribe kit (New England Biolab [NEB]) or an in house chemically defined protocol (not published). A typical in vitro transcription (IVT) reaction consists of a linearised DNA template, multiple enzyme components including T7 polymerase, and nucleotide triphosphates (NTPs), wherein mRNA was co-transcriptionally capped with a 5'cap analogue.^{36,37} At least one modified NTP was applied in the examples provided here. The template DNA was digested using DNase before purification of the crude IVT mixture.

2.3 | mRNA purification

Each chain of the encoded mAb (HC, LC) shared the general process scheme—filtration (impurity clearance), chromatography (polishing), filtration (formulation)—though their operation differed for each molecule's chromatography step. All buffers used for mRNA purification were prepared with non-DEPC treated Nuclease Free water (Thermo Fisher Scientific) and 0.22 µm filtered before use.

2.3.1 | TFF development

TFF was selected as first step in mRNA purification to both remove IVT related process impurities and condition the mRNA sample for chromatography. Initial TFF process development was carried out using purified mRNA sample (mAb 1 HC and LC) on a system comprising Masterflex tubing, a Watson Marlow 100 series feed pump, and Pendotech sanitary pressure sensors. Water permeability was checked before and after experimentation. The running parameters were carried through to a KrosFlo® KR2i (Repligen) TFF system equipped with an mPES 100 kDa MWCO hollow fiber (HF) membrane (Repligen).

2.3.2 | Chromatography steps

In order to separate HC and LC mRNA from the remaining impurities in the IVT reaction, such as T7 polymerase and residual nucleotides, resin screen experiment was performed. CaptoCore 400 (Cytiva), CHT Ceramic Hydroxyapatite Type II (Biorad) and CIMmultus™ Oligo-dT Monolith (BIA separations) were most promising resins hence further investigation was performed.

Ceramic hydroxyapatite type II (CHT type II)

CHT Type II was chosen to purify LC Gradient elution (5–500 mM sodium phosphate buffer, 100 mM NaCl, 2 mM CaCl₂ at pH 6.5, 7.0, 7.5) was first performed to identify the optimal pH. Second, wash and elution condition was designed based on gradient results, respectively.

Capto™ Core 400

Multimodal Capto™ Core 400 column (Cytiva) was operated under condition that consist of equilibration (50 mM Tris, 300 mM KCl, pH 8.0), re-equilibration, and elution (50 mM Tris, 4 M NaCl, pH 7.4), and then sanitized with 1 M NaOH and 30% 1-propanol.

Affinity capture evaluation

Two affinity resins with a poly-adenylated tail were evaluated among to replace TFF, thusto streamlining the manufacturing process. POROS™ Oligo (dT)25 (Thermo Fisher Scientific) and CIMmultus™ Oligo-dT Monolith (BIA separations) were used to purify both HC and LC that follows the bind-and-eluate mechanism following recommended and optimized condition.

2.4 | mRNA quantification

2.4.1 | RiboGreen™ RNA assay

Quant-iT™ RiboGreen™ RNA Assay Kit (ThermoFisher Scientific) was used to quantify mRNA during process development using RNase-free vials. Fluorescence of 1xTE diluted samples were read in triplicate and compared with readings from purified luciferase L7202 mRNA

standard (Trilink Biotechnologies) using an Envision plate reader (PerkinElmer) with excitation/emission wavelengths 485/520 nm.

2.4.2 | UV spectroscopy

Absorbance measurements at 260 and 280 nm were carried out on a Trinean Lunatic (Unchained Labs) using a dual pathlength microfluidic chip to assist in-process measurement.

2.5 | Reversed phase HPLC

In-process mRNA samples were analyzed using ion paired reversed phase high-performance liquid chromatography (RP-HPLC) to track impurities and measure mRNA integrity. The method analyses 10 µg of mRNA loaded on a hydrophobic interaction analytical column (Agilent Technologies PL1912-3803) connected to the 1260 Infinity II analytical Agilent HPLC system. Triethylammonium acetate buffer was used as an ion pairing reagent, with 50% acetonitrile as the organic phase. This method was adapted from RNA separation methods documented by Welch et al.³⁸ The test samples were run in sequence with a purified mRNA standard encoding the same IgG construct to validate target molecule retention time.

2.6 | Capillary electrophoresis sodium dodecyl sulfate (CE-SDS)

A Bioanalyzer 2100 (Agilent) loaded with an RNA Pico 6000 or Protein 230 microfluidic chip was used for in-process sample testing. Manufacturer specifications were followed for measurement of diluted samples (protein samples were non-reduced). Fragment analyzer (FACE) 5400 (Agilent) was also used as capillary electrophoresis sodium dodecyl sulfate (CE-SDS) method to analyze mRNA quality post purification.

2.7 | Residual total protein

Residual total protein was measured using NanoOrange® Protein Quantitation (Invitrogen) following a custom sample pre-treatment and read on SpectraMax M3 plate reader (Molecular Devices). Samples were pre-diluted to nucleic acid concentration of 150 µg/ml into 1X NanoOrange working solution prepared per vendor's direction. All samples (with and without 3 µg/ml BSA spike), standards, and negative control (1x Nano Orange working solution alone) were subjected to 250 U Benzonase incubation on shaking at 500 rpm for 1 h at room temperature. Following benzonase treatment, all samples, standards, and controls were heated at 95°C for 30 min and cooled to room temperature. Total protein concentrations was interpolated from the standard curve.

2.8 | ds mRNA assay

Homogeneous Time-Resolved Fluorescence (HTRF[®]) technology was used to measure double-stranded RNA (ds mRNA), which is a quantitative and more sensitive method compared with traditional dot blot. The modified and non-modified IVT mRNA samples were measured by using the commercial ds mRNA FRET assay kit (Perkin Elmer-Cis Bio, 64RNAPEG). The HTRF based kit is intended for the simple and rapid quantification of viral dsRNA in cell lysates of mammalian cells infected by a virus and offers a simple alternative to quantitative PCR (qPCR). The detection principle of this kit is based on (Homogeneous Time-Resolved Fluorescence). dsRNA is detected in a sandwich assay by using anti dsRNA antibody labeled with Europium cryptate (donor), and anti dsRNA antibody labeled with d2 (acceptor). When the dyes are in close proximity, the excitation of the donor with a light source (laser or flash lamp) triggers a fluorescence resonance energy transfer (FRET) toward the acceptor, which in turn fluoresces at a specific wavelength (665 nm).

2.9 | THP-1 dual reporter cell-based assay

THP1-Dual[™] cells (NF-KB-SEAP and IRF-Lucia Luciferase reporter monocytes) were purchased from InvivoGen. THP1-Dual cells have functioning ssRNA—and dsRNA-sensing innate immune pathways.³⁹ Cells were cultured in Gibco[®] RPMI 1640 medium (Thermo Fisher Scientific) supplemented with 10% Foetal Bovine Serum (Heat Inactivated Gamma Irradiated, Gibco), 10 µg/ml blasticidin, and 100 µg/ml zeocin; 100,000 cells were transfected in triplicates with 20 ng mRNA using Lipofectamine MessengerMAX (Invitrogen) accordingly to manufacturer's instructions. The plate was then incubated for 24 h at 37°C, 5% CO₂ in the cell incubator. Subsequently, 20 µl of cell culture supernatants were mixed with 50 µl of QUANTI-Luc[™] assay solution (InvivoGen) and luminescence was immediately measured using luminescence microplate reader (Envision[™]). The results were collected and analyzed as fold change measurement over the untreated cells.

3 | RESULTS AND DISCUSSION

3.1 | DNA template purification

Prior to mRNA synthesis in IVT, linearised *E. coli* plasmid DNA must be purified. Phenol/chloroform extraction and spin columns are commonly used to purify linearised DNA for the subsequent IVT though scalability for this approach remains challenging. We compared the resulting purity of mRNA following IVT and a one-step hydrophobic interaction chromatography (HIC) purification whose IVT template, linearised DNA that was purified by one of four scalable strategies side-by-side with phenol chloroform purification as a reference. Using bind and elute mode, anion chromatography media, Q Sepharose HP, was considered for separating target linearized DNA from other contaminants, including enzymes as well as larger genomic DNA and

plasmid DNA. PlasmidSelect Xtra was evaluated in flowthrough chromatography mode, where target linearised DNA remained in the flow-through, and contaminants were captured on the resin, which offers the benefits of high productivity when compared with bind and elute chromatography mode. As indicated in Table 1, when we evaluated mRNA quality data post IVT and HIC, linearised DNA purified by Q Sepharose HP followed by TFF produced the best quality of mRNA among all the tested conditions with 90% of the product in the main peak. Surprisingly, TFF alone demonstrated the lowest mRNA product quality post IVT, which suggests further studies may be necessary to study the membrane type and associated MWCO. Though the purification strategy developed for HC, it was likewise applicable for LC molecules. Overall compared with phenol chloroform, Sepharose-based strong anion exchange chromatography generated substantially comparable mRNA product quality and was selected for all subsequent development to prepare linearised DNA template for IVT.

Linearized DNA serves template from which mRNA is synthesized, containing the polymerase promoter, coding region, poly-A tail as well as essential elements for *E. coli* growth. It is linearized from circular plasmid in a defined reaction of enzymes and reaction buffer. Phenol/chloroform extraction is a widely adopted method to purify linearised DNA for research purpose as well as in industrial settings at large scale, the step can become onerous. Furthermore, though it is possible to scale up the reaction, it requires management of large quantities of hazardous chemicals which is at odds with the preferred use of “generally regarded as safe” (GRAS) chemicals in manufacturing.

While a single step of ultrafiltration for buffer exchange was successfully applied to purify linearised DNA from the mixture of plasmid DNA and enzymes by others.³¹ In our hands, the ultrafiltration without additional chromatography generated low monomer purity mRNA. Ultimately, AEX chromatography followed by ultrafiltration yielded the highest purity mRNA of the methods investigated. The process difference could be due to the properties of the specific linearized DNA template sequence design and the restriction enzymes used in the DNA linearization step. In addition, the AEX removes impurities that can potentially exhibits negative impact on mRNA quality during IVT reaction, which is worthwhile to investigate further.

3.2 | mRNA purification

Toward the goal of developing a scalable and GMP compatible purification scheme for mRNA, below we present a series of studies on purification unit operations applied to mRNA. The model mRNA studied below encodes either a heavy chain (HC, ~2 kb) or light chain (LC, ~1 kb) component of a monoclonal antibody.

3.2.1 | TFF development

Crude IVT product consists of various components in addition to target mRNA, including for example T7 RNA polymerase, unused 5' cap analogue, residual DNA template, other residual proteins, and free

TABLE 1 HC mRNA quality comparison utilizing different purification strategies of linearized DNA post IVT by FACE

pDNA purification method	mRNA quality post IVT ^a			
	%HMW	%LMW	%MP	%total area
No purification applied	10.1	7.2	82.8	100.1
TFF only	6.0	19.4	74.7	100.1
Q sepharose HP and TFF	3.7	6.2	90.1	100.0
PlasmidSelectXtra and TFF	8.0	6.2	85.8	100.0
phenol/chloroform load (control)	3.5	5.4	91.0	99.9

^aHMW, high molecular weight species; LMW, low molecular weight species; MP, main product.

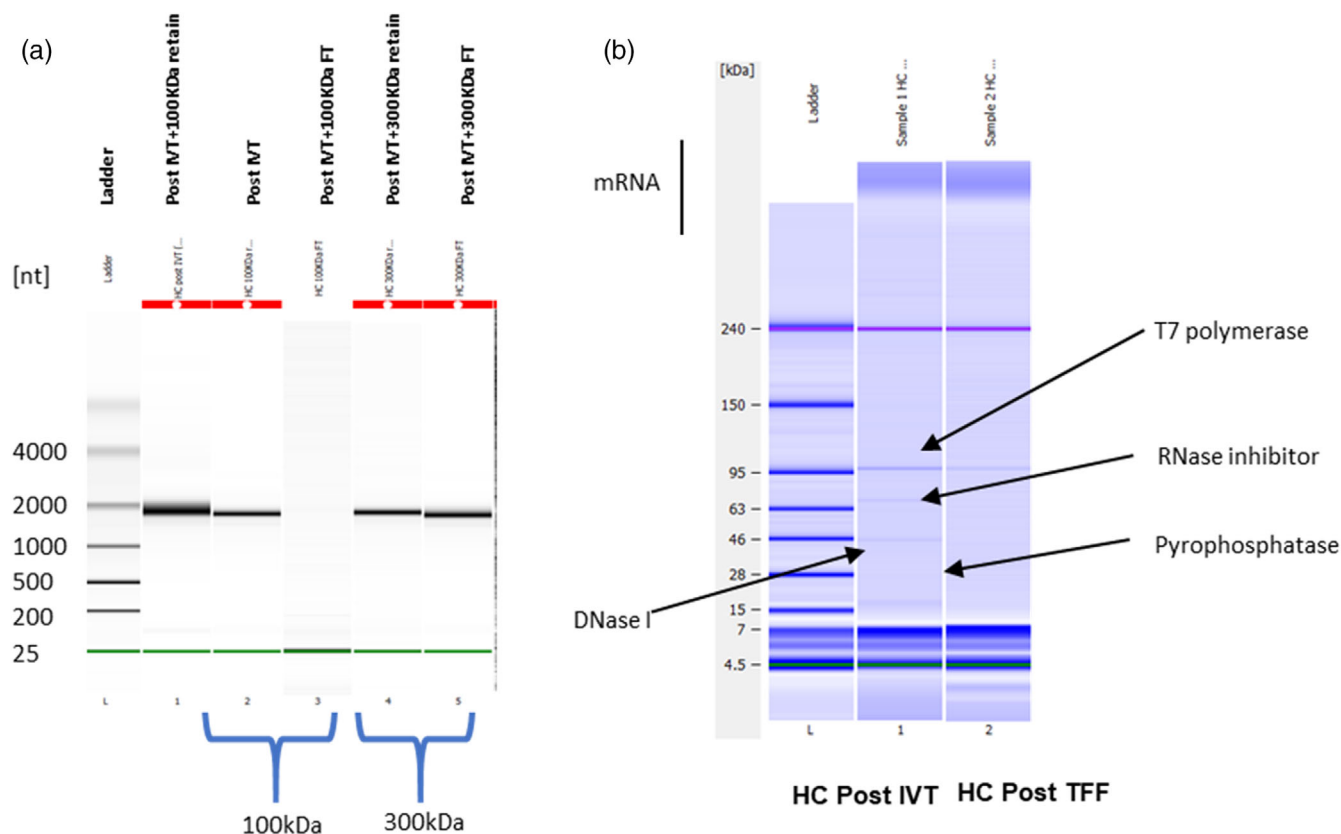


FIGURE 1 (a) CE-SDS of mAb 1 HC mRNA samples before and after UF/DF using 100 kDa and 300 kDa PES membrane respectively. (b) Non-reducing CE to monitor impurities clearance. it was observed that post UF/DF, all the enzymes used in IVT were substantially cleared except for T7 polymerase

nucleotides. TFF was originally chosen as first step not only to remove low molecular species but also because it serves to condition the mRNA load into a buffer suitable for the subsequent chromatography step.

To determine the required MWCO of the membrane, mRNA purity and integrity was monitored by CE-SDS after 100 or 300 kDa PES membrane filtration. Based on the mRNA product retention, which was 50% higher in the case of the 100 kDa membrane, it was selected as the appropriate MWCO for TFF. Additional monitoring of product and process related impurities were carried out using non-reducing CE (Figure 1). We observed good clearance for the majority of impurities with T7 polymerase being the only protein impurity remaining at detectable levels.

With the membrane material and MWCO defined, further process development was performed to set the operation parameters of TFF step following standard methods.⁴⁰ Flux vs transmembrane pressure (TMP) excursion curves were generated for purified HC mRNA, diluted to 1.1 mg/ml (Figure 2). Nuclease-free water permeability tests were carried out before and after each run. Low TMP values (≤ 5 psi) are preferred as flux saturated beyond that point for flow rates between 8.1 to 23 ml/min; 5 psi was selected for operation. For long mRNA (10,000 nt), other work reported TMP set to 2 psi and a recommended shear rate range of $500\text{--}5000\text{ s}^{-1}$ ⁴¹ encompassing the maximum shear rate (1930 s^{-1} ; 23 ml/min) tested in this work; 23 ml/min was selected as the feed flow rate for both HC and LC.

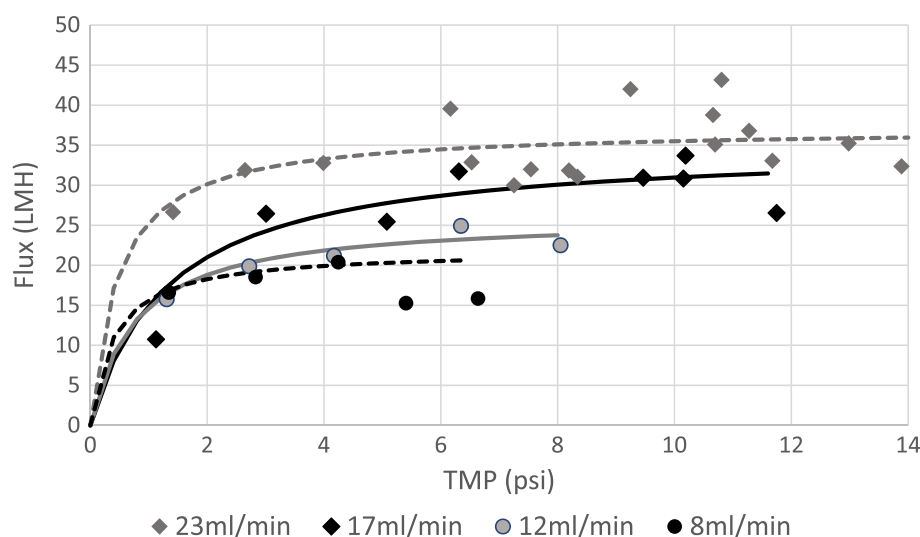


FIGURE 2 TMP Excursion curves for the mAb 1 HC molecule at various flowrates and fit with a saturation curve-based model

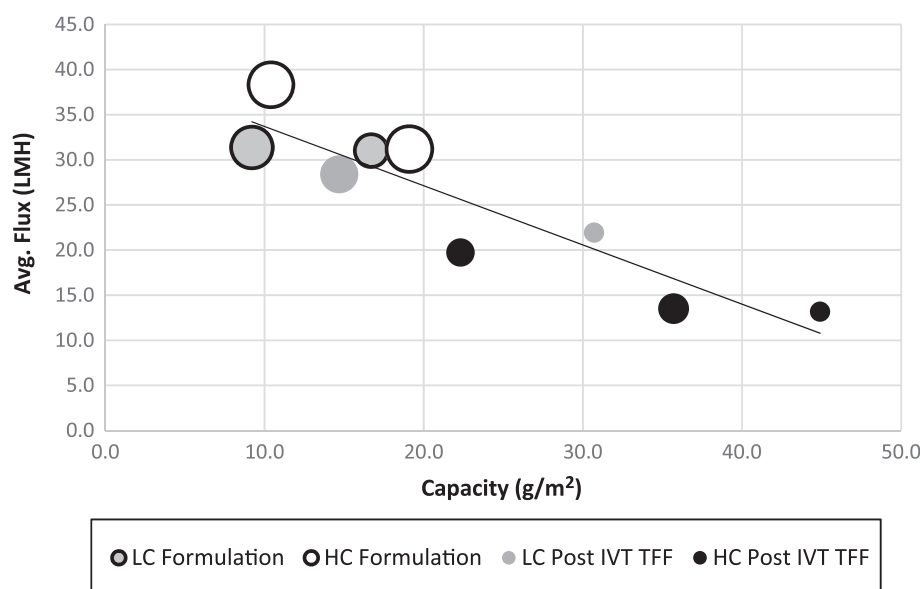


FIGURE 3 Average permeate flux (LMH) vs membrane load capacity for crude IVT product and formulation for distinct ($n = 5$) mRNA molecules, of which two are described mAb1 and mAb2 and the rest of mRNA molecules encoding other mAbs. Bubble size represents % mass recovery ranging from 60% to 96% as computed from UV A_{260} readings before and after TFF. For crude IVT product, RNA purification via spin column was performed before UV reading to avoid confounding the measurement

To select an appropriate concentration operating range, permeate flux was measured as a function of mRNA concentration. TMP was stabilized in full recycle mode at 23 ml/min for at least 2 min at each sampling point before flux was measured. The impact of mRNA concentration on flux was also investigated, 1 mg/ml is preferred considering that the high flux (>25 LMH) can be achieved (Data S1). Generally a weak decay in flux was observed (Data S1) with a slight plateau suggesting operation below 1.5 mg/ml, and preferably 1 mg/ml favoring higher flux >25 LMH for both LC and HC encoding mRNA molecules. Filtration extent was set to at least six diafiltration volumes (6DV) based on strong reductions of A_{260} absorbing species in the filtrate (see Data S1).

As the mass of mRNA loaded onto the membrane is increased, a decreased flux—related, in part, to higher associated impurity load—and RNA recovery must be considered. Visualizing this balance, a bubble plot (Figure 3) below shows a roughly linear decrease in flux with respect to capacity regardless of purification

stages (diafiltration or formulation) or molecule sequences ($n = 5$; four of which encode mAb1 and mAb 2). The concentration of material during diafiltration was 1.0 ± 0.2 mg/ml, the target set during development. Fitting a line ($R = 0.826$) to the whole dataset, 46 g/m² can be computed to represent the capacity corresponding to a 70% flux decay compared with the flux achieved at 10 g/m². For clarification of crude IVT material, other work loaded 25–65 g/m² RNA on an equivalent filter material and MWCO (mPES, 100 kDa) membrane cassette for 10DV using filtered water with resulting permeate flux of 20–120 Lm⁻² h⁻¹ albeit with higher TMP (1 bar, 14.5 psi) and comparable or lower feed flowrate (300–1050 LMH feed flux).⁴² The increase in flux may be due to the combination of membrane format, RNA concentration, and or increased feed pressure which, for the hollow fiber membrane in this work was limited to <5 psi otherwise fouling was observed, even at 1 mg RNA/ml. Differences in IVT content may also play a role in performance differences.

While Funkner et al. apply relatively higher-pressure filtration for a 500–5000 kb preferred length molecule, at large scale (120 mg RNA), Scorza et al. apply a gentler technique for a 10 kb mRNA molecule. A low TMP (2 psi), low feed flux (~ 150 LMH), low mass load (~ 1.5 g/m²) approach is carried out on a higher MWCO (500 kDa) hollow fiber mPES membrane using 8 DV of 10 mM Tris, 250–500 mM NaCl or KCl, pH 8.0 to achieve high purity (>95%) and recovery (>80%).⁴¹ A side by side comparison of such techniques on the same starting material may elucidate a preferred operation, though at face value, large molecules likely need gentler processing.

TFF recovery during formulation after chromatography polishing was reliably high although its lower range was comparable with the best recovery observed for post IVT TFF equal to ca. 80%. Better recoveries are generally observed when less material is loaded onto the membrane while filtration duration increased with capacity. Based on these data, we preferred loading at lower capacities, at or below 30 g/m², that also enables step completion within a working day.

TFF is widely used for sample concentration and buffer exchange in protein manufacturing. Its application is widespread across several modalities such as viral purification⁴³ and plasmid DNA purification⁴⁴ and appropriate membrane pore size and operating parameters are often determined empirically. Many mRNA manufacturing patents claim at least one filtration step of TFF.^{31,41,42} Taken together, while lower TMP and membrane load capacity is required to avoid fouling when purifying mRNA compared with mAbs (typically ca. TMP 20 psig; 1000 g/m² capacity for formulation), permeate flux is generally comparable or somewhat increased. TFF following the IVT reaction is effective in removing protein residuals, which was demonstrated from CE and residual host cell protein assay. It also improves performance of the subsequent chromatography step.

In an innovative application of TFF, Sanofi (Translate Bio) describes filtration embodiments with a 500MWCO, 75 cm² membrane loaded with a mixture of 2.2 ml crude IVT product.³⁵ If IVT yield is assumed to be 5 mg/ml, the corresponding capacity of RNA precipitated on the membrane would be of ~ 1.5 g/m², lower than those capacities explored in this work and others which could be appropriate for their use of TFF for *in situ* precipitation rather than extensive diafiltration.

3.2.2 | Chromatography development for HC and LC mRNA

While small sized impurities such as RNA NTPs, 5'cap analogue reagent, and enzymes that are present in crude IVT product are substantially cleared during TFF, T7 polymerase (99 kDa) (Figure 1) and undigested large DNA fragments may be retained. As a result, an orthogonal polishing step was desirable to obtain high-purity mRNA.

An initial resin screen data suggested ceramic hydroxyapatite has the potential for acceptable mRNA purification. Hydroxyapatite contains two types of sites for binding interactions: positively charged calcium and negatively charged phosphate groups. In this case, calcium affinity occurs through interaction between phosphate groups

on mRNA and carboxyl clusters distributed on the crystal structure of the matrix. The binding can be further enhanced by the addition of CaCl₂.

However, electrostatic repulsion between immobilized phosphate groups and mRNA phosphates reduces the net binding energy. Therefore, NaCl together with CaCl₂ were added to enable the binding. The mRNA product binds to the column while the process-related impurities are removed in the non-bound or wash fractions. pH 7.0 was determined as optimal pH to run on CHT column (see Data S1). As shown in Figure 4, there were multiple peaks observed during gradient elution (5–500 mM sodium phosphate buffer) suggesting potential separation between the target mRNA molecule and impurities. Indeed, CE-SDS revealed that there was no HC mRNA molecule detected in the flowthrough fractions (A2, A6, A10) or pre-peaks fractions (C5, C6) while the electropherogram indicates that impurities can be potentially removed in the flow through and pre-peaks before elution. Subsequently, an isocratic elution was designed and optimized to enable large-scale manufacturing, replacing gradient elution. The running condition is suitable for both HC and LC mRNA. Experiments where impurities were deliberately added to the chromatography load suggested that T7 polymerase was present in the flowthrough (A2–A10), whereas DNA fragments in the washes. The impact of pH on mRNA separation with CHT type II were also studied, pH 7.0 was optimal under tested condition (see Data S1).

Though CHT type II column is suitable for both HC and LC mRNA in principle, an alternative resin (Capto™ Core400) was selected to further improve the recovery of the heavy chain encoding mRNA species compared with CHT type II. Capto™ Core resin is operated in flow-through mode. The separation mechanism is hierarchical, operating first by size exclusion restricting access to the pores for large species (e.g., mRNA), followed by chemical interaction (AEX/hydrophobic interaction) once small impurities have penetrated the neutral shell's pores (400 kDa MWCO for Capto™ Core400 or 700 kDa for the Capto™ Core700 resin).

Chromatography samples were prepared by loading diafiltered IVT product up to a target capacity of 1.0 mg RNA/ml resin. Additionally, a series of impurity spike experiments were carried out to understand the species' separation behavior during chromatography, shown in Figure 5 below.

Residual template DNA is digested by DNase added to the crude IVT product and expected to be cleared by TFF however, if DNase I digestion is incomplete, lengthy fragments that persist after TFF may be too large to be cleared by the column. To investigate this, we loaded both intact and partially digested pDNA at a mass commensurate with expected impurity levels for an RNA load capacity of 1 mg RNA/ml resin, equal to 12.2 μ g DNA. A 0.8% agarose gel (see Data S1) shows the intact and digested pDNA (1/10 enzyme load). As expected based on size, undigested linear DNA template is co-eluted at low levels with product in the flow-through evidenced by a slight rise in the purple trace in the first phase, while digested DNA template is cleared during the high-salt elution (Step 3: elution, green trace). Because un-digested DNA can be co-eluted, attention must be taken to ensure upstream DNase digestion is complete.

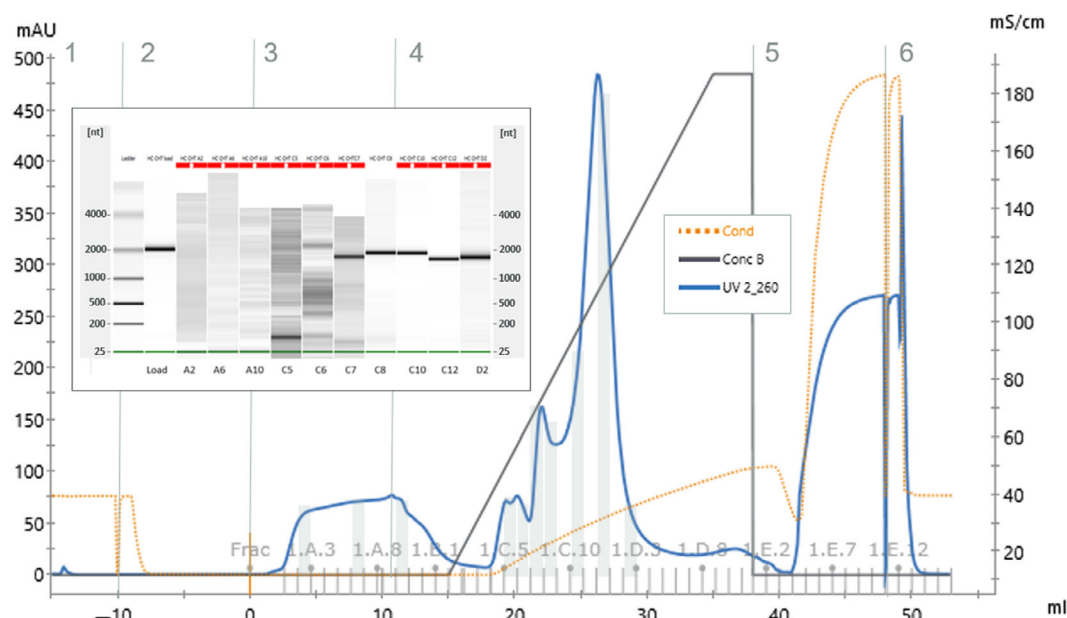


FIGURE 4 UV absorbance and conductivity profile of gradient elution (5–500 mM NaPO₄, 100 mM NaCl, 2 mM CaCl₂, pH 7.0) of mAb1 HC mRNA with CHT type II column using 1 ml/min. Chromatogram phases are as follows: 1. pre-equilibration, 2. equilibration, 3. sample load, 4. elution, 5. sanitization, 6. regeneration. Highlighted chromatogram fractions (gray bars) and sample load were analyzed by capillary electrophoresis (inset)

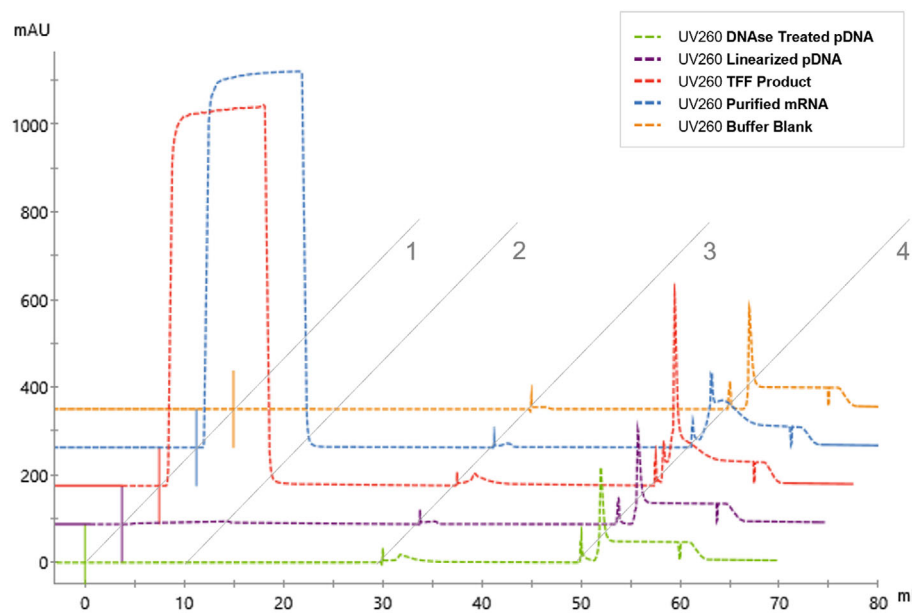


FIGURE 5 Five overlaid Core 400 Chromatograms showing the TFF product trace (red), purified RNA product (blue), digested linearized DNA (1/10 enzyme load DNase; green), and undigested linearized pDNA loaded to typical masses expected from an average yield IVT reaction. A buffer-reference trace are shown in orange since the sanitization step shows absorbance at 260 nm. Method Steps 1–4; 1. flow-through; 2. re-equilibration; 3. elution; 4. sanitization are indicated

Notably, some mRNA is lost on the column; overall recovery using 300 cm/h is $80 \pm 4\%$; $n = 3$. Comparing the sanitization phase traces in Figure 5 of both pure mRNA and IVT product loads, the broadness of the shoulder on the second peak suggests some mRNA appears to be entering the pores and retained on the column until sanitization. Decreasing the linear velocity below 200–300 cm/h appears to increase mRNA losses (see Data S1). The linear velocity set point of 300 cm/h is consistent with other work reporting operating at 275 cm/h on a Capto™ Core700 for sample volumes up to 1 L.³² In their reported example settings, a column with lower aspect ratio (length/diameter,

$L/d = 4.1$) is used with a higher load 2.0 mg RNA/ml resin which may represent advantages compared with this work using a narrower column of the same volume ($L/d = 20$) and half the loaded capacity. Considering our hypothesis that mRNA is being partially retained as it enters the pores slightly, a lower aspect ratio column reduces residence time—time mRNA is in contact with the porous shell—reducing the chance of loss via pore penetration and therefore potentially enhancing recovery. Finally, impurity load must also be considered.

Unlike HC mRNA, the conditions developed here were not applicable for LC mRNA that is presumed to have penetrated the pores of

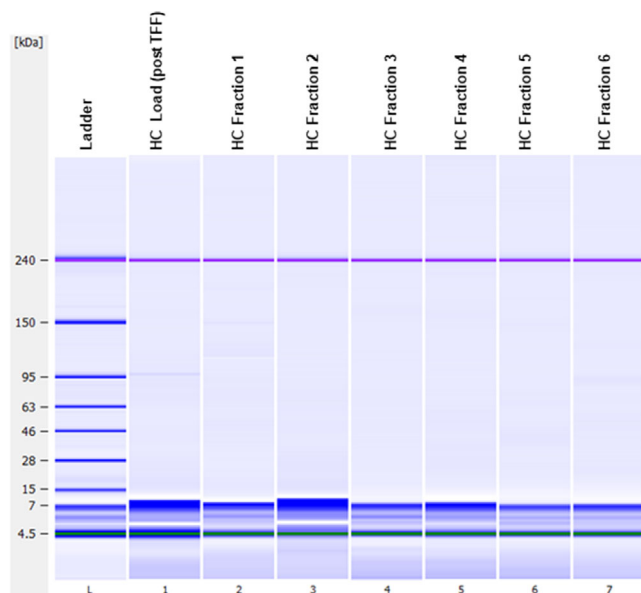


FIGURE 6 NR (non-reducing) SDS-CE of Capto™ core 400 in-process samples from mRNA encoding mAb1 HC

Capto™ Core 400 based on its poor recovery (see Data S1). In addition to mRNA,^{31,32} Capto™ Core technology has been applied to purification of extracellular vesicles⁴⁵ and virus purification⁴⁶ whose nonlinear morphology may be better suited to this technology.

After polishing, the heavy chain coding mRNA should be free of dsRNA and protein, specifically T7 polymerase (99 kDa) which may not optimally be cleared during the initial pre-column 6 DV of 100 kDa MWCO TFF. CE-SDS was used to measure residual proteins after Capto™ Core 400 purification. It was demonstrated in Figure 6 that the residual T7 was no longer observed in different fractions collected from the Capto™ Core 400 flow through stream.

3.3 | Process integration and impurity clearance

A representative, scalable, end-to-end process, which comprises two filtration steps bracketing a single chromatography polishing step was applied and tailored to each of the two pairs of (HC and LC) mRNA molecules. Tables 2 to 5 summarize the process performance and quality attributes from those two processes. Product recoveries are reported in Tables 2 and 3 for each sequential process step during linearised pDNA and mRNA purification for HC and LC molecules encoding mAb 1 and mAb 2. The overall process performance is comparable between mAb 1 and mAb 2 (i.e., within HC or LC classes of similar length), however, between HC and LC molecules, the average overall recovery of HC mRNA is higher, 58% compared with 48% for LC mRNA. The total recoveries calculated based on absorbance (A_{260}) and fluorescence-based Ribogreen™ are comparable though differences in orthogonal quantitation observed between steps varies, which suggests that the residual impurities (e.g., nucleic acid monomers and DNA template) potentially interfere with the readings using

absorbance and kit-based assays. Next, key process attributes: fragment, monomer purity, and dsRNA content are tabulated for the four molecules (see Tables 4 and 5). Residual dsRNA and protein levels are similarly low for all processes. While RP-HPLC and FACE comparably report purity values >89% for all molecules, FACE tends to report slightly lower main peak purity. Both capping efficiency and the length of polyA tail are within expectation (data not reported).

Evaluating dsRNA level and immunogenicity is critical to enable mRNA use as therapeutics and vaccines. It was previously reported that the potential residual dsRNA generated in the IVT reaction can trigger an innate immune response.⁴⁷ As a result, we have also introduced dsRNA assay testing and immunogenicity assay as part of mRNA quality control monitoring. The results suggest that our mRNA process generated limited amount of dsRNA (Figure 7) at less than 1 ng/mg. Negligible elicitation of immune response was observed, evidenced by a lack of enhanced interferon IFN signals resulting from exposure of cells to the mRNA product (Figure 8). While the mechanism is still being investigated, we believe that the in-house IVT reaction conditions limit the generation of dsRNA. The process capability to clear dsRNA is still under investigation.

3.4 | Use of Oligo-dT resin as an alternative capture step

Affinity chromatography appears better suited for scale-up of bulk purification of crude IVT compared with the TFF operation described above, based on its enhanced recovery and faster operation. To streamline process purification, implementation of affinity chromatography was investigated for both HC and LC encoding mRNA molecules by assessing dynamic binding capacity (DBC) for the Oligo-dT POROS™ Oligo (dT)25 and CIMmultus™ Oligo-dT Monolith columns. Each was launched to effectively capture mRNA through the polyadenylated mRNA tail and we compared their performance to capture both HC and LC encoding mAb 1 respectively (Table 6). CIMmultus™ Oligo-dT Monolith showed significantly higher recovery compared with POROS™ Oligo (dT). Both resins led to good product quality as demonstrated on RP-HPLC, where 90% mRNA purity was achieved. DBC was determined and 3 mg/ml was achieved with 90% recovery for the monolithic column whereas the porous bead linked oligo-dT resin had up to 50% reduction in load capacity for the longer HC molecule. In our experience, recovery varies depending on the length of mRNA. Therefore, it is important to measure DBC empirically for mRNA products of interest. Overall, compared with TFF, a ~30% enhancement in recovery is observed when switching to affinity chromatography as capture step.

Recently, manufacturing methods of mRNA are fast evolving. Early practical methods of mRNA purification like ion-paired reversed phase HPLC using linear gradient elution were reported by Kariko, et al.¹⁷ In the reported method, both triethylammonium acetate (TEAA) and acetonitrile were applied and it demonstrates that the HPLC method successfully removed dsRNA which was identified to be responsible for innate immune activation. On the other hand,

TABLE 2 HC mRNA Integrated process performance

STEP	Species	mAb1 % recovery (A260)	mAb1 %recovery (Ribogreen)	mAb2 % recovery (A ₂₆₀) ^a	mAb2 %recovery (Ribogreen)
Chromatography (Q sepharose HP)	Linearized DNA	88	n/a	75	n/a
TFF (100 kDa PES)		76	n/a	67	n/a
DNA Overall		67	n/a	50	n/a
In vitro transcription of mRNA					
TFF (diafiltration; 100 kDa PES)	mRNA	78	99	73	91
Capto™ Core 400		76	57	85	67
TFF (Formulation; 100 kDa PES)		98	100	97	100
RNA Overall		58	56	60	61

^aValues are computed by spectrophotometry at 260 nm. For TFF following IVT, Input mass and the resulting recovery is computed from A260 following spin column filtration.

TABLE 3 LC mRNA Integrated process performance

STEP	Species	mAb1 % recovery (A260)	mAb1 %recovery (Ribogreen)	mAb2 %recovery (A ₂₆₀) ^a	mAb2 %recovery (Ribogreen)
Chromatography (Q sepharose HP)	Linearised DNA	70	n/a	86	n/a
TFF (100 kDa PES)		67	n/a	54	n/a
DNA Overall		47	n/a	46	n/a
In vitro transcription of mRNA					
TFF (diafiltration; 100 kDa PES)	mRNA	70	75	60	48
CHT ceramic hydroxyapatite type II		81	73	93	75
TFF (formulation; 100 kDa PES)		95	100	80	100
RNA overall		54	55	45	36

^aValues are computed by spectrophotometry at 260 nm. For TFF following IVT, Input mass and the resulting recovery is computed from A260 following spin column filtration.

TABLE 4 Quality attributes of mRNAs encoding mAb 1 purified from integrated process

Key analytical attributes	mAb 1 HC ^a			mAb 1 LC		
	%LMW	%MP	%HMW	%LMW	%MP	%HMW
FACE	5.3	91.1	3.5	8.3	90.6	1.3
RP-HPLC	8	92	ND	6.8	93.2	ND
dsRNA (ng/ml)	0.65			0.57		
residual total protein (ng/ml)	<0.5			<0.5		

^aHMW, high molecular weight species; LMW, low molecular weight species; MP, main product.

TABLE 5 Product quality of mRNAs encoding mAb 2 purified from integrated process

Key analytical attributes	%LMW	mAb 2 HC ^a		mAb 2 LC		
		%MP	%HMW	%LMW	%MP	%HMW
FACE	6.5	92	3.5	8.5	89.4	2.2
RP-HPLC	1.9	98	ND	4.5	93	ND
dsRNA (ng/ml)		0.47			0.59	
Residual total protein(ng/ml)		<0.5			<0.5	

^aHMW, high molecular weight species; LMW, low molecular weight species; MP, main product.

Moderna Therapeutics has previously patented mRNA purification processes using ion exchange chromatography, where mRNA is required to be heated during manufacturing.⁴⁸ Both organic solvent and sample heating are not favored operations during mRNA large-scale manufacturing. Organic solvents can pose storage, disposal, and handling issues whereas, increasing the sample temperature can

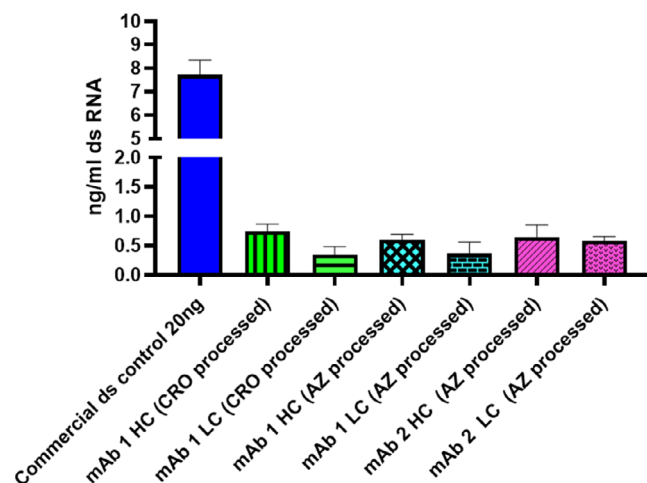


FIGURE 7 dsRNA quantity in each sample was measured by using anti-dsRNA Eu Cryptate antibody that was provided by the viral double-stranded RNA detection kit (Perkin Elmer-Cis-Bio). Total 20 ng of IVT mRNA was measured in each well and the samples were read in triplicates. Double stranded quantity of in house purified heavy and the light chain of the mRNA measured less than 1 ng/ml. This amount was comparable to the quantity of ds mRNA of the commercially purified samples

potentially require enhanced system control during chromatography or else be subject to additional sources of process variability.

During downstream process development, molecules' properties such as charge and molecular weight are often two key parameters to be considered since they may impact higher order structure alongside chemical interaction. AEX is often utilized in nucleic acid purification as mRNA is negatively charged however, we often experienced low recovery of AEX resins, therefore, we focused on size and molecular weight differences to separate small impurities from the much larger mRNA product. When considering TFF, RNA are less structured compared with proteins and can potentially suffer from RNA precipitation and denaturation.⁴⁹ Furthermore, the effect of mRNA sequence on secondary structure is not fully understood and so overall, special attention should be paid to membrane pore sizes and operation parameters in order to design suitable purification processes.

TFF has several advantages for RNA purification⁴¹ and is also applied in virus and DNA processing.⁴⁴ For example, once an appropriate membrane size is selected, it can potentially remove substantial amount of impurities involved in the IVT reaction, as a result, a more platform friendly approach can be achieved using TFF as first step post IVT. However, TFF tends to be lengthy and mRNA losses can exceed 30% in this study. The new introduction of Oligo dT affinity chromatography facilitates the selective and direct capture of mRNA to separate enzymes and other defined reagents used in IVT.

Still, it is worthwhile to be aware of the effect of mRNA length on process performance and the likelihood of retaining residual DNA fragments impurities, which can interact with mRNA or the column and still requires clearance. For example, the DBC corresponding to

FIGURE 8 IFN response was measured for both mAb 1 (HC and LC) and mAb 2 (HC and LC) in THP1-Dual reporter cells transfected with 20 ng of IVT mRNA at 24 h after transfection. In house purified mRNA samples did not induced IFN response. Poly I:C (positive control) significantly induced IFN activation whereas CleanCap Firefly Luciferase 5-methoxyuridine (negative control) remained silent. Data represent two independent experiments with $n = 3$ technical replicates

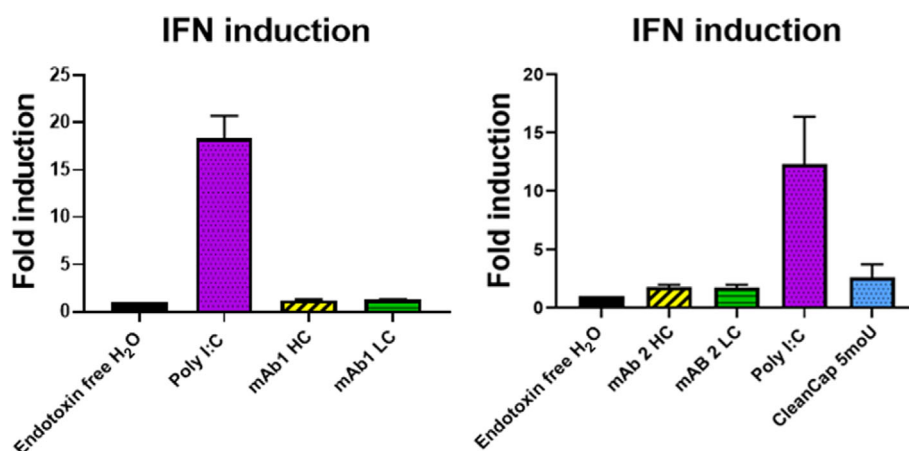


TABLE 6 Oligo-dT performance comparison using both HC and LC mRNA that encodes mAb1

	CHT type II LC	POROS™ Oligo (dT)25		CIMmultus™ Oligo-dT Monolith	
		HC	LC	HC	LC
Dynamic Binding capacity(mg/ml)(10% breakthrough)	0.5	1.5	2.5	3	3
% Recovery	90	66	61	95	96
% Purity (RP-HPLC)	94	91	96	97	98

5% breakthrough can double for an mRNA 1 kb long compared with 3 kb though capacities for longer mRNA can benefit from increased residence times and loading concentrations.⁵⁰ Salt concentration and identity can be tuned to increase DBC, being mindful to avoid precipitation which may also be a function of sequence make-up.⁵¹ Although Oligo dT chromatography resin manufacturers recommend optimization based on the size of mRNAs, it nevertheless offers a great opportunity to be included in mRNA production and purification platforms. Model-assisted process development has been applied in other biological products to help with process understanding and identification of key quality attributes.^{52–54} Therefore, mechanistic and statistical modeling could likewise be considered to accelerate mRNA process development and shorten the timeline between research, development, and products' launch. Recent advances in process analytics⁵⁵ may also be applied in mRNA manufacturing to understand mRNA's process in real-time starting from IVT to drug substance.

4 | CONCLUSIONS

Here we presented a process as a series of case studies that were successfully integrated and applied to multiple mRNA molecules separately encoding the HC and LC of two monoclonal antibodies. Upon co-formulation and delivery, potency was shown for the intact antibody with a negligible immune response, owing, in part, to low dsRNA amount present in the purified mRNA.

The need for tailoring of polishing chromatography to the mRNA's length, and to an unknown degree its sequence, is a remaining challenge. Here, the smaller LC coding mRNA was tailored to the CHT column where the HC coding molecule, only ca. 2000 nt long, had a severely reduced recovery. Conversely, the HC coding mRNA had improved recovery on the CaptoCore400 where the ca. 1000 nt had poor recovery. Oligonucleotide Poly-T based affinity resins can partly dampen the dependence of load capacity on length compared with surface binding or mixed-mode based SEC resins. Future work will be focused on increasing their load capacity.

TFF is a powerful tool to purify mRNA, especially with its dual functionality as an impurity removal tool and a load conditioning step for chromatography. It delivers adequate recovery and purity during diafiltration for buffer exchange and bulk impurity clearance albeit more gently, with lower TMP and capacity yet only moderately less flux than a standard antibody processes. Suitable membranes and their operation conditions from mRNA at various length requires further studies.

Overall, as new robust tools are developed, refined, and better understood, we expect that robust molecule agnostic platform methods will be developed for generating large quantities of pure, non-immunogenic therapeutic material to treat disease and fulfill the promise of mRNA advances.

AUTHOR CONTRIBUTIONS

Tingting Cui and Kareem Fakhfakh designed the experiments and performed mRNA process development and process integration; Hannah

Turney contributed to affinity capture evaluation, development and reversed phase HPLC; Gülin Güler-Gane contributed to dsRNA mRNA assay; Aleksandra Toloczko contributed to THP-1 Dual reporter cell based assay. Tingting Cui and Kareem Fakhfakh wrote the manuscript together with Hannah Turney, Gülin Güler-Gane, and Aleksandra Toloczko. Tingting Cui, Martyn Hulley, and Richard Turner supervised the research work.

ACKNOWLEDGMENTS

We are grateful for the support from DARPA. The authors would like to acknowledge Maiken Kristiansen and Alexandra Ristani from Cell Culture and Fermentation Sciences for providing IVT material and Sara Trabulo, Alistair Hines, Shirley Lee, Stephany Staniforth, Abigail Markle from Analytical Sciences to provide advice and support molecule and impurity analysis during process integration. We are grateful also to our colleagues from Discovery Sciences for early work pioneering mRNA molecule design and pDNA team to supply plasmid DNA. The authors would also like to thank William Wang and Luis Santos for reviewing the manuscript and providing valuable suggestions.

FUNDING INFORMATION

This research was developed with funding from the Defense Advanced Research Projects Agency (DAPRA) under HRO11-18-3-001. The views, opinions, and/or findings expressed are those of the authors and should not be interpreted.

CONFLICT OF INTEREST

There is no conflict interest.

PEER REVIEW

The peer review history for this article is available at <https://publons.com/publon/10.1002/btpr.3301>.

DATA AVAILABILITY STATEMENT

All the data that have been included in the article is available in the public domain.

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REFERENCES

1. Kübler H, Maurer T, Stenzl A, et al. Final analysis of a phase I/IIa study with CV9103, an intradermally administered prostate cancer immunotherapy based on self-adjuvanted mRNA. *J Clin Oncol*. 2011;29(15_suppl):4535.
2. Robinson E, MacDonald KD, Slaughter K, et al. Lipid nanoparticle-delivered chemically modified mRNA restores chloride secretion in cystic fibrosis. *Mol Ther*. 2018;26(8):2034–2046.
3. Truong B, Allegri G, Liu XB, et al. Lipid nanoparticle-targeted mRNA therapy as a treatment for the inherited metabolic liver disorder arginase deficiency. *Proc Natl Acad Sci U S A*. 2019;116(42):21150–21159.
4. Chow MYT, Qiu Y, Lam JKW. Inhaled RNA therapy: from promise to reality. *Trends Pharmacol Sci*. 2020;41(10):715–729.
5. Patel AK, Kaczmarek JC, Bose S, et al. Inhaled nanoformulated mRNA polyplexes for protein production in lung epithelium. *Adv Mater*. 2019;31(8):e1805116.

6. Bartley JM, Cadar AN, Martin DE. Better, faster, stronger: mRNA vaccines show promise for influenza vaccination in older adults. *Immunol Investig*. 2021;50(7):810-820.
7. Jackson NAC, Kester KE, Casimiro D, Gurunathan S, DeRosa F. The promise of mRNA vaccines: a biotech and industrial perspective. *NPJ Vaccines*. 2020;5:11.
8. Xie W, Chen B, Wong J. Evolution of the market for mRNA technology. *Nat Rev Drug Discov*. 2021;20(10):735-736.
9. Tugues S, Burkhard SH, Ohs I, et al. New insights into IL-12-mediated tumor suppression. *Cell Death Differ*. 2015;22(2):237-246.
10. Rabinovich PM, Komarovskaya ME, Ye ZJ, et al. Synthetic messenger RNA as a tool for gene therapy. *Hum Gene Ther*. 2006;17(10):1027-1035.
11. Damase TR, Sukhovshin R, Boada C, Taraballi F, Pettigrew RI, Cooke JP. The limitless future of RNA therapeutics. *Front Bioeng Biotechnol*. 2021;9:628137.
12. Wang D, Tai PWL, Gao G. Adeno-associated virus vector as a platform for gene therapy delivery. *Nat Rev Drug Discov*. 2019;18(5):358-378.
13. Choi A, Koch M, Wu K, et al. Safety and immunogenicity of SARS-CoV-2 variant mRNA vaccine boosters in healthy adults: an interim analysis. *Nat Med*. 2021;27(11):2025-2031.
14. Bueno SM, Abarca K, Gonzalez PA, et al. Safety and immunogenicity of an inactivated SARS-CoV-2 vaccine in a subgroup of healthy adults in Chile. *Clin Infect Dis*. 2021;75(1):e792-e804.
15. Sahin U, Karikó K, Türeci Ö. mRNA-based therapeutics—developing a new class of drugs. *Nat Rev Drug Discov*. 2014;13(10):759-780.
16. Pardi N, Hogan MJ, Weissman D. Recent advances in mRNA vaccine technology. *Curr Opin Immunol*. 2020;65:14-20.
17. Kariko K, Muramatsu H, Ludwig J, Weissman D. Generating the optimal mRNA for therapy: HPLC purification eliminates immune activation and improves translation of nucleoside-modified, protein-encoding mRNA. *Nucleic Acids Res*. 2011;39(21):e142.
18. Baierdorfer M, Boros G, Muramatsu H, et al. A facile method for the removal of dsRNA contaminant from in vitro-transcribed mRNA. *Mol Ther Nucleic Acids*. 2019;15:26-35.
19. Pardi N, Muramatsu H, Weissman D, Kariko K. In vitro transcription of long RNA containing modified nucleosides. *Methods Mol Biol*. 2013;969:29-42.
20. Chen R, Wang SK, Belk JA, et al. Engineering circular RNA for enhanced protein production. *Nat Biotechnol*. 2022.
21. Taylor PC, Adams AC, Hufford MM, de la Torre I, Winthrop K, Gottlieb RL. Neutralizing monoclonal antibodies for treatment of COVID-19. *Nat Rev Immunol*. 2021;21(6):382-393.
22. Barone PW, Wiebe ME, Leung JC, et al. Viral contamination in biological manufacture and implications for emerging therapies. *Nat Biotechnol*. 2020;38(5):563-572.
23. Sergeeva OV, Kotliansky VE, Zatselin TS. mRNA-based therapeutics – advances and perspectives. *Biochemistry (Mosc)*. 2016;81(7):709-722.
24. Sahin U, Kariko K, Türeci O. mRNA-based therapeutics—developing a new class of drugs. *Nat Rev Drug Discov*. 2014;13(10):759-780.
25. Van Hoecke L, Roose K. How mRNA therapeutics are entering the monoclonal antibody field. *J Transl Med*. 2019;17(1):54.
26. Rathore AS, Winkle H. Quality by design for biopharmaceuticals. *Nat Biotechnol*. 2009;27(1):26-34.
27. Leppek K, Byeon GW, Kladwang W, et al. Combinatorial optimization of mRNA structure, stability, and translation for RNA-based therapeutics. *bioRxiv* 2021.
28. Hartmann G. Nucleic acid immunity. *Adv Immunol*. 2017;133:121-169.
29. Koubek J, Lin KF, Chen YR, Cheng RP, Huang JJ. Strong anion-exchange fast performance liquid chromatography as a versatile tool for preparation and purification of RNA produced by in vitro transcription. *RNA*. 2013;19(10):1449-1459.
30. Easton LE, Shibata Y, Lukavsky PJ. Rapid, nondenaturing RNA purification using weak anion-exchange fast performance liquid chromatography. *RNA*. 2010;16(3):647-653.
31. von der MÜlbe F, Reidel L, Ketterer T, et al., Method for Producing RNA. US Patent US 10,017,826 B2. February 15, 2016. 2018.
32. Berlanda Scorza F, Wen Y, Geall A, Porter F et al. Inventors; GLAXOSMITHKLINE BIOLOGICALS SA, assignee. RNA purification methods. US Patent EP 2,970,948 B1. 2018.
33. Hoge S, Issa W, Miracco Edward J, Nelson J, Rabideau Amy E, Butora G, Inventors; MODERNATX INC, assignee. High Purity RNA Compositions and Methods for Preparation Thereof. US Patent US10,653,712 B2. June 6, 2018. 2020.
34. Gagnon P, Goričar B, Peršič S, Černigoj U, Štrancar AJCGTI. Two new capture options for improved purification of large mRNA. *Cell gene ther insights*. 2020;6:1035-1046.
35. Abysalhi J, Vargas J, Derosa F, Anusha D, Inventors; MODERNATX INC, assignee. Methods and Compositions for Messenger RNA Purification. US Patent WO 2020/097509 A1. November 8, 2019. 2020.
36. Milligan JF, Uhlenbeck OC. Synthesis of small RNAs using T7 RNA polymerase. *Methods Enzymol*. 1989;180:51-62.
37. Brunelle JL, Green R. In vitro transcription from plasmid or PCR-amplified DNA. *Methods Enzymol*. 2013;530:101-114.
38. Biba M, Welch CJ, Foley JP. Investigation of a new core-shell particle column for ion-pair reversed-phase liquid chromatography analysis of oligonucleotides. *J Pharm Biomed Anal*. 2014;96:54-57.
39. Nelson J, Sorensen EW, Mintri S, et al. Impact of mRNA chemistry and manufacturing process on innate immune activation. *Sci Adv*. 2020;6(26):eaz6893.
40. Millipore E. A Hands-On Guide to Ultrafiltration/Diafiltration Optimization Using Pellicon® Cassettes. EMD Millipore Corporation; 2013.
41. Berlanda Scorza F, Wen Y, Geall A, Porter F. Inventors; GLAXOSMITHKLINE BIOLOGICALS SA, assignee. RNA Purification Methods. US Patent US 2021/0214388 A1. January 25, 2021. 2021.
42. Funkner A, Dorner S, Sewing S, et al., Inventors. WO/2016/193206 A method for producing and purifying RNA, comprising at least one step of tangential flow filtration. 2016.
43. Grzenia DL, Carlson JO, Czermak P, Han B, Specht RK, Wickramasinghe SR. Purification of densovirus by tangential flow ultrafiltration. *Biotechnol Prog*. 2006;22(5):1346-1353.
44. Nunes C, Sousa A, Nunes JC, Morao AM, Sousa F, Queiroz JA. Supercoiled plasmid DNA purification by integrating membrane technology with a monolithic chromatography. *J Sep Sci*. 2014;37(11):1229-1236.
45. McNamara RP, Caro-Vegas CP, Costantini LM, et al. Large-scale, cross-flow based isolation of highly pure and endocytosis-competent extracellular vesicles. *J Extracell Vesicles*. 2018;7(1):1541396.
46. Tseng Y-F, Weng T-C, Lai C-C, Chen P-L, Lee M-S, Hu AY-C. A fast and efficient purification platform for cell-based influenza viruses by flow-through chromatography. *Vaccine*. 2018;36(22):3146-3152.
47. Mu X, Greenwald E, Ahmad S, Hur S. An origin of the immunogenicity of in vitro transcribed RNA. *Nucleic Acids Res*. 2018;46(10):5239-5249.
48. Issa WJ, Barberio JL, Aunins JG, Afeyan N, B, Inventors; Moderna Therapeutics, Inc., assignee. Ion Exchange Purification of mRNA. US patent WO2014144767A1 2014.
49. Geiger J, Trembl M, Inventors; Ethris GmbH, assignee. mRNA purification by tangential flow filtration. US patent WO2020165158A1 2020.
50. Flook K. Supporting development of mRNA-based therapies by addressing large-scale purification challenges. *Cell gene ther insights*. 2021;7(5):489-502.
51. Mencin N, Krušić A, Ličen J, et al. Increasing dynamic binding capacity of oligo(dT) for mRNA purification: experimental results using CIM 96-well plates. *Bioprocess Int*. 2022.
52. Narayanan H, Luna MF, von Stosch M, et al. Bioprocessing in the digital age: the role of process models. *Biotechnol J*. 2020;15(1):e1900172.
53. Shekhawat LK, Rathore AS. An overview of mechanistic modeling of liquid chromatography. *Prep Biochem Biotechnol*. 2019;49(6):623-638.
54. van de Berg D, Kis Z, Behmer CF, et al. Quality by design modelling to support rapid RNA vaccine production against emerging infectious diseases. *NPJ Vaccines*. 2021;6(1):65.

55. Armstrong A, Horry K, Cui T, et al. Advanced control strategies for bioprocess chromatography: challenges and opportunities for intensified processes and next generation products. *J Chromatogr A*. 2021; 1639:461914.

SUPPORTING INFORMATION

Additional supporting information can be found online in the Supporting Information section at the end of this article.

How to cite this article: Cui T, Fakhfakh K, Turney H, et al. Comprehensive studies on building a scalable downstream process for mRNAs to enable mRNA therapeutics. *Biotechnol. Prog.* 2022:e3301. doi:[10.1002/btpr.3301](https://doi.org/10.1002/btpr.3301)